Matrix operations: exercises

28-E01 Two triangular substitutions

Solve A=[1 2;3 4], b=(5,11)^{T} using unpermuted LU. Give L, U, y, x and check the original equations. Why does a zero first pivot in A=[0 1;1 0] not imply noninvertibility?

28-E02 Schur complement and inverse

View A=[1 1;2 5] as four scalar blocks. Compute S and the block inverse, then verify every AA^{-1} entry. Can the formula be used directly when A is invertible but chosen leading B singular? Give an example.

28-E03 A line fit with nonzero residual

For(0,0),(1,2),(2,1) and $\hat{y}=c+mt$, give design matrix, Gram, right side, coefficients, predictions, and residuals. Verify $A^{T}r=0$ and squared error. Replacing data by(0,1),(1,2),(2,3), does the design rank decrease?

28-E04 Later permutations must swap earlier L entries

For A=[0 2 1;1 1 0;2 3 4], b=(0,0,7)^{T}, trace partial pivoting, both row swaps, p, multipliers, L, U, and solution. If swap two omits the saved first-column multipliers, which product row already differs from PA?

28-E05 Both complexity reductions

For T=[I B 0;0 I C;0 0 I], verify inverse[I −B BC;0 I −C;0 0 I] without assuming B, C commute. Explain assumptions $I(3n)=O(I(n))$, $I(n)=\Omega(n^2)$. Expand $Tinv(n)\leq 2Tinv(n/2)+O(M(n))$ under $M(n/2)\leq cM(n)$, $c<1/2$. How does general real nonsingular A reduce to SPD inversion?

28-E06 SPD and Cholesky

Compute the positive-diagonal Cholesky factor of A=[4 2 2;2 5 1;2 1 3] entry by entry and verify the product. Compute and prove positivity of the first scalar-pivot Schur complement. Show why positive original diagonals are insufficient using a counterexample vector for[1 2;2 1].

Matrix operations: exercises (continued)

28-E07 Rank deficiency and error spaces

For A=[1 1;1 1;1 1], b=(0,2,1)^{T}, find every least-squares coefficient vector, the unique prediction, and residual. Prove moving away from that prediction increases squared error. Compare κ_2 for diag(1,10^{-3}) and its Gram, distinguishing sensitivity from arithmetic running time.

28-E08 One vector in two coordinate systems

Use A=[1 0;1 1;1 2], b=(0,2,1)^{T}, p=(1/2,1,3/2)^{T}, r=(−1/2,1,−1/2)^{T}. Compute u=p/||p|| and v=r/||r|| and verify orthonormality. Find the coordinates of b in section E=span{p,r} using basis (u,v), then reconstruct its R^3 coordinates. Prove E∩Col(A)=span{p}. Why does the horizontal line in this section not make Col(A) one-dimensional?

> Complete the derivation before consulting the separate solutions PDF.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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